Experiment 7

1-Bit Full Subtractor and 2:4 Decoder

Aim: To design and implement 1-bit Subtractor and 2:4 Decoder using Behavioral Modelling and the results through simulation.

Apparatus: Vivado Tool.

(i)1-bit Full Subtractor using Behavioral Modelling

Theory:

A **Full Subtractor** is a combinational circuit that takes three inputs: Minuend (A), Subtrahend (B) and Borrow-in (Bin), and produces two Outputs: Difference (D) and Borrow-Out (Bout).

Verilog Code:

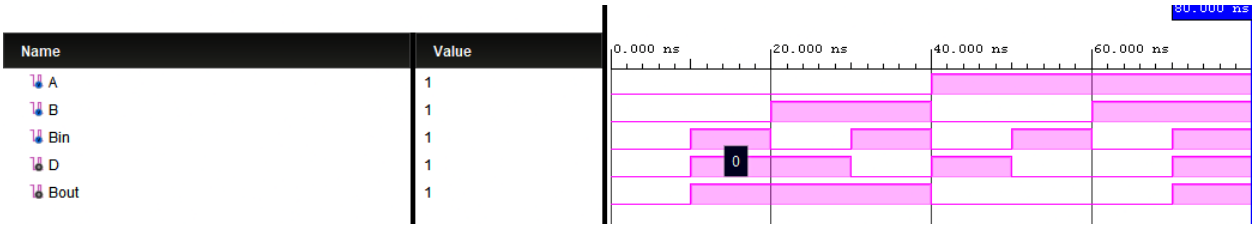
```
module full_subtractor(
    input A, B, Bin,
    output reg D, Bout
);

    always @ (A or B or Bin) begin
        D    = A ^ B ^ Bin;
        Bout = (~A & B) | (~(A ^ B) & Bin);
    end

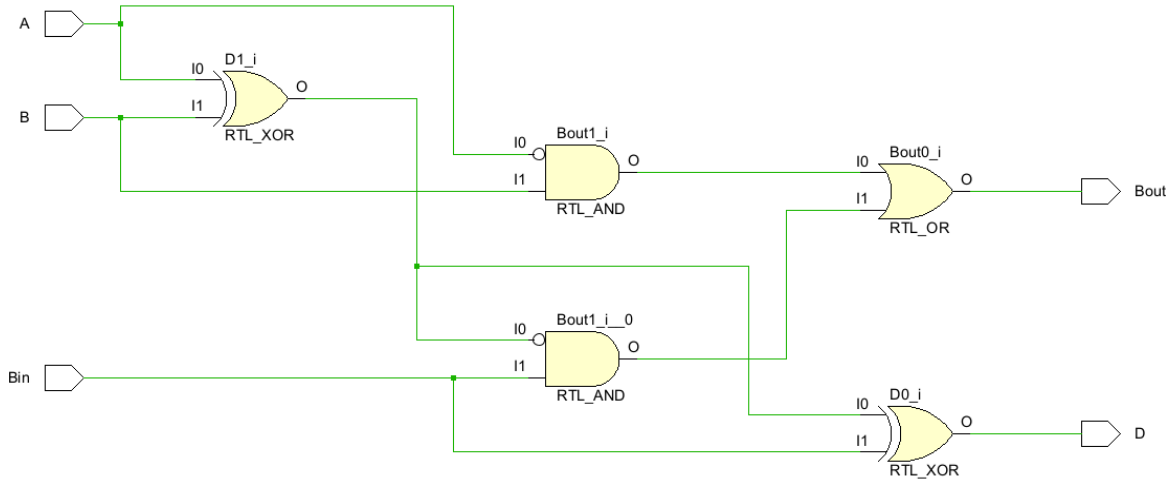
endmodule
```

Difference is generated by XORing the inputs A, B, Bin, representing the subtraction logic. **Borrow** is generated by examining the conditions when borrow occurs: when Minuend (A) is
smaller than Subtrahend (B) or when the borrow propagates.

Simulation:



Schematic:



(ii) 2:4 Decoder using Behavioral Modelling

Theory:

A 2:4 Decoder is a combinational circuit that decodes a 2-bit input (S0, S1) into 4 outputs (Y0, Y1, Y2, Y3). The output corresponding to the binary value of the input is set to 1, while the rest remain 0.

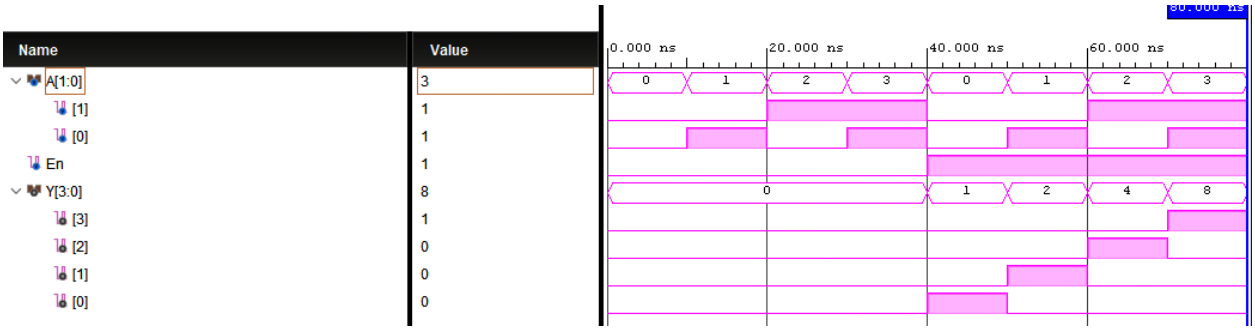
Verilog Code:

```
module decoder_2to4(
    input [1:0] A,
    input En,
    output reg [3:0] Y
);

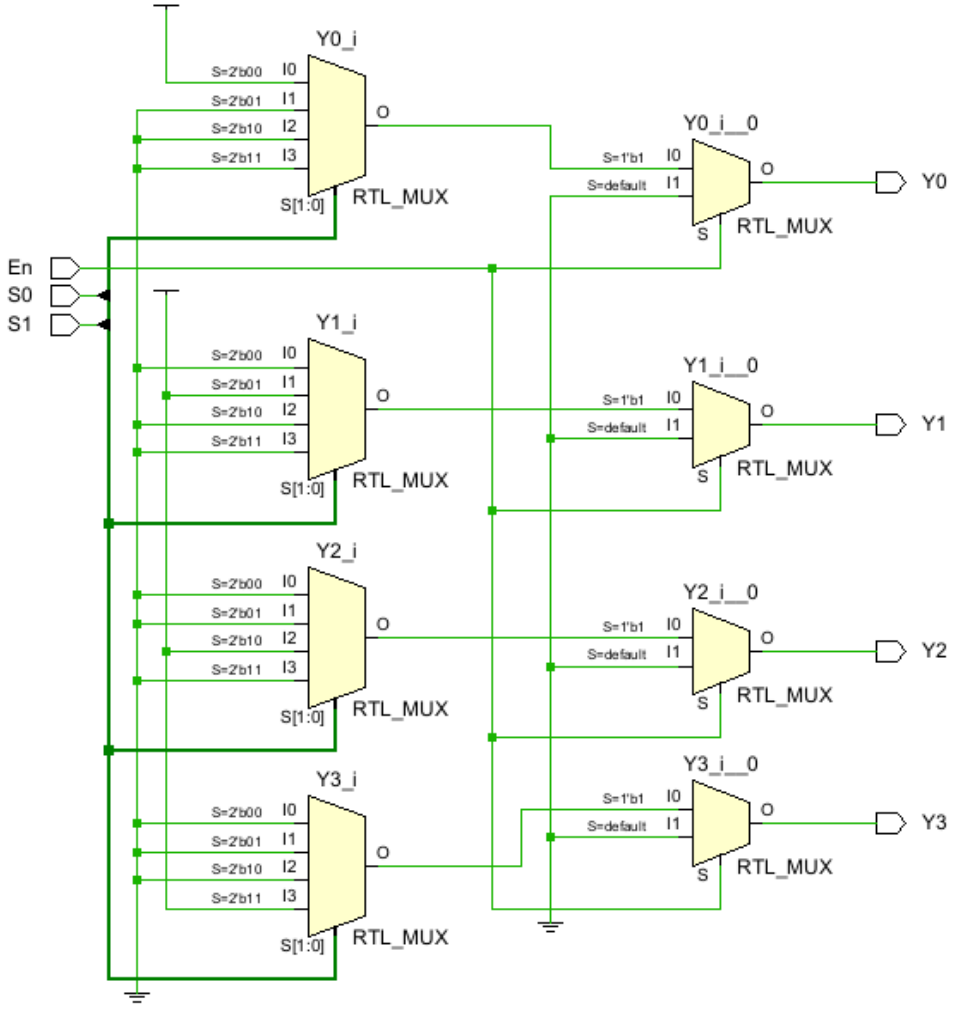
    always @ (A or En) begin
        if (En) begin
            case (A)
                2'b00: Y = 4'b0001;
                2'b01: Y = 4'b0010;
                2'b10: Y = 4'b0100;
                2'b11: Y = 4'b1000;
            endcase
        end else begin
            Y = 4'b0000;
        end
    end

endmodule
```

Simulation:



Schematic:


Y0_i
S=2'b00
S=2'b01
S=2'b10
S=2'b11
I0
I1
I2
I3
O
S[1:0]
RTL_MUX
Y0_i__0
S=1'b1
S=default
S
Y0
En
S0
S1
Y1_i
Y1_i__0
Y1
Y2_i
Y2_i__0
Y2
Y3_i
Y3_i__0
Y3

Procedure

1. Open **Vivado Design Suite** and create a new RTL project by specifying the project name, location, and required FPGA device.
2. Create a **Verilog source file** for the 1-bit full subtractor and define the inputs **A, B, Bin** and outputs **D, Bout**.
3. Implement the full subtractor using **Behavioral Modelling** with an always block to generate the Difference and Borrow-out outputs.
4. Create a **testbench** for the full subtractor and apply the required combinations of A, B, and Bin inputs.
5. Run **Behavioral Simulation** in Vivado and observe the waveform to verify the Difference and Borrow-out outputs.
6. Create a Verilog source file for the **2:4 Decoder** with a 2-bit input, enable input, and four outputs **Y0, Y1, Y2, and Y3**.
7. Implement the 2:4 decoder using **Behavioral Modelling**, using a case statement to select the corresponding output based on the input combination when Enable is active.
8. Create a **testbench** for the decoder and apply different input and Enable combinations to verify its operation.
9. Run **Behavioral Simulation** and observe the waveform and RTL schematic to verify that the selected decoder output becomes HIGH while the remaining outputs remain LOW.
10. Verify the simulation results of both circuits against their respective truth tables and record the simulation waveforms and schematics.

Conclusion

The **1-bit Full Subtractor and 2:4 Decoder** were successfully designed and implemented using **Behavioral Modelling in Vivado**. The simulation results verified the correct generation of Difference and Borrow-out for the full subtractor and the correct selection of decoder outputs for the applied input combinations.